

Telco Customer Churn Analysis

Business Analyst Project Report

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Submission Date:

19th May 2026

1. Problem Statement

Customer churn is a major challenge for telecom companies because losing customers directly impacts revenue and profitability. Understanding why customers leave the service is essential for improving customer retention and business performance.

The objective of this project is to analyse customer churn behaviour, identifying the key factors influencing churn, segment customers based on churn risk, and provide business recommendations to reduce customer attrition and improve customer satisfaction.

2. Dataset Description

The Telco Customer Churn dataset contains customer demographic information, telecom service details, billing information, contract types, payment methods, and customer churn status.

The dataset is used to analyse customer behaviour and identify the major factors contributing to customer churn.

Dataset Information

Total Records: 7043

Total Features: 21

Key Features Included

Customer demographics (gender, senior citizen, partner, dependents)

Service details (internet service, phone service, streaming services)

Contract information

Payment methods

Monthly and total charges

Customer tenure

Churn status

The dataset contains both categorical and numerical variables which are useful for exploratory data analysis and churn prediction.

3. Approach / Methodology

The project followed a structured data analysis approach to identify customer churn patterns and business insights.

Steps Followed

Data Loading and Inspection

- Imported the dataset using Python libraries such as Pandas.
- Performed initial exploration using functions like head(), shape() and info().

Data Cleaning and Preprocessing

- Handled missing values in the dataset.
- Standardized column names and categorical values.
- Converted data types where necessary.

Exploratory Data Analysis (EDA)

- Analysed customer churn distribution.
- Compared churn behaviour across contract types, payment methods, and monthly charges.
- Created visualizations using Matplotlib and Seaborn.

KPI Analysis

- Calculated important business metrics such as:
- Churn Rate
- Average Revenue Per User (ARPU)

Customer Segmentation

- Identified high-risk customers likely to churn.
- Identified loyal customers based on tenure and service behaviour.

Basic Churn Prediction Model (Bonus)

- Built a Logistic Regression model to predict customer churn.
- Evaluated model performance using accuracy score.

Business Insights and Recommendations

- Derived key insights from the analysis.
- Suggested retention strategies to reduce customer churn.

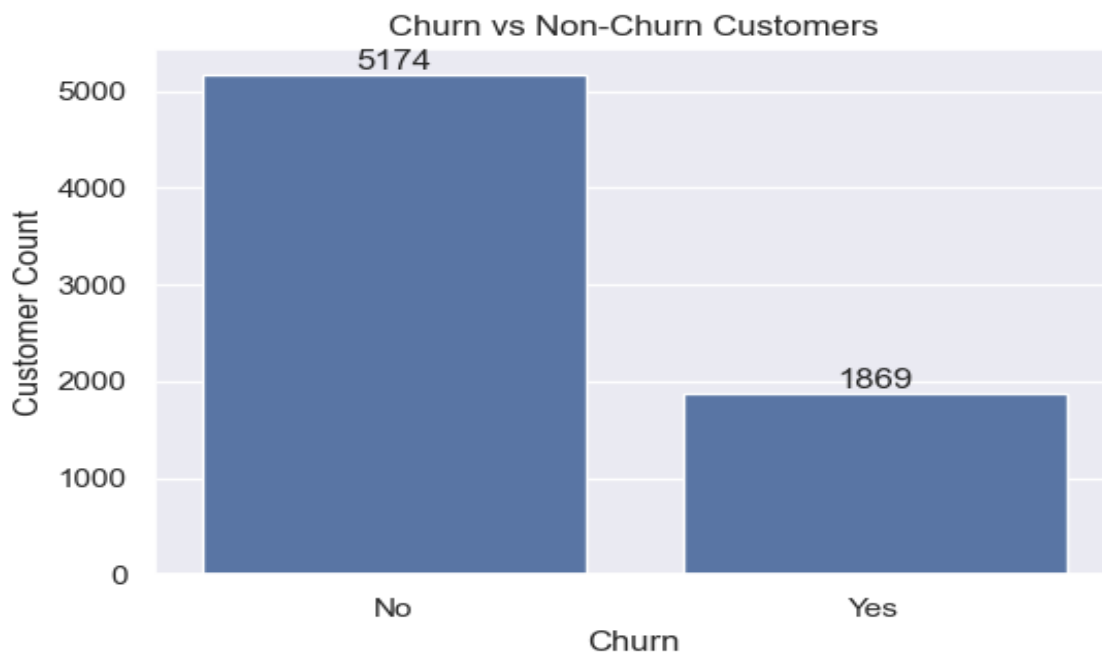
4. Analysis / Model

Exploratory Data Analysis (EDA) was performed to identify churn patterns. A Logistic Regression model achieved approximately 81.48% prediction accuracy.

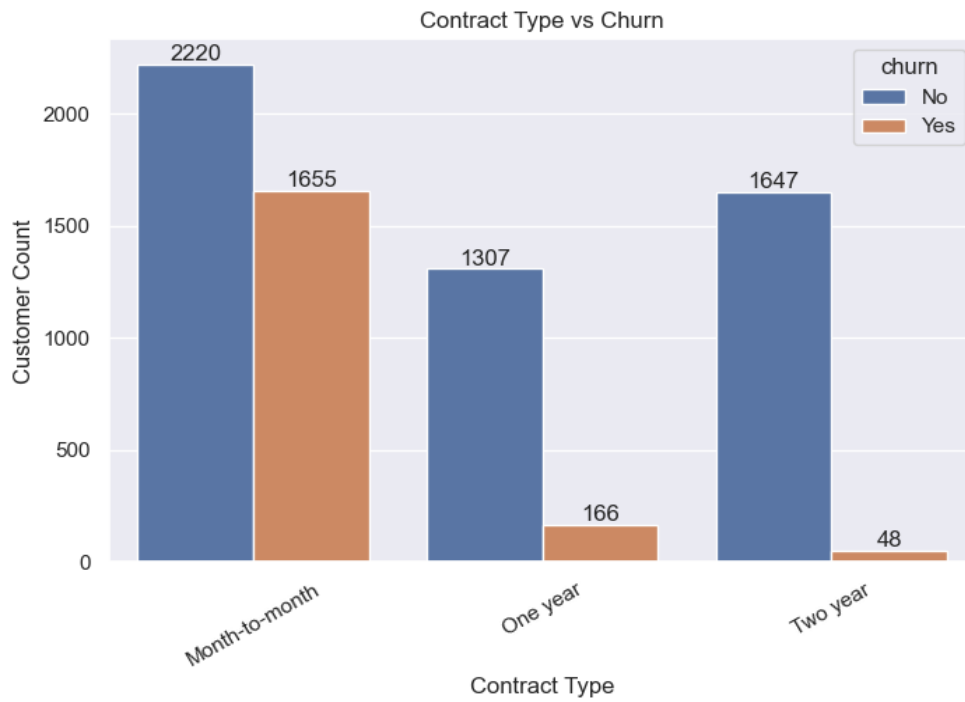
5. Key Insights

- Approximately 26.54% customers churned from the telecom service.
- Customers with month-to-month contracts have the highest churn rate.
- Higher monthly charges increase churn probability.
- Long-tenure customers are more loyal and less likely to churn.
- The churn prediction model achieved 81.48% accuracy.

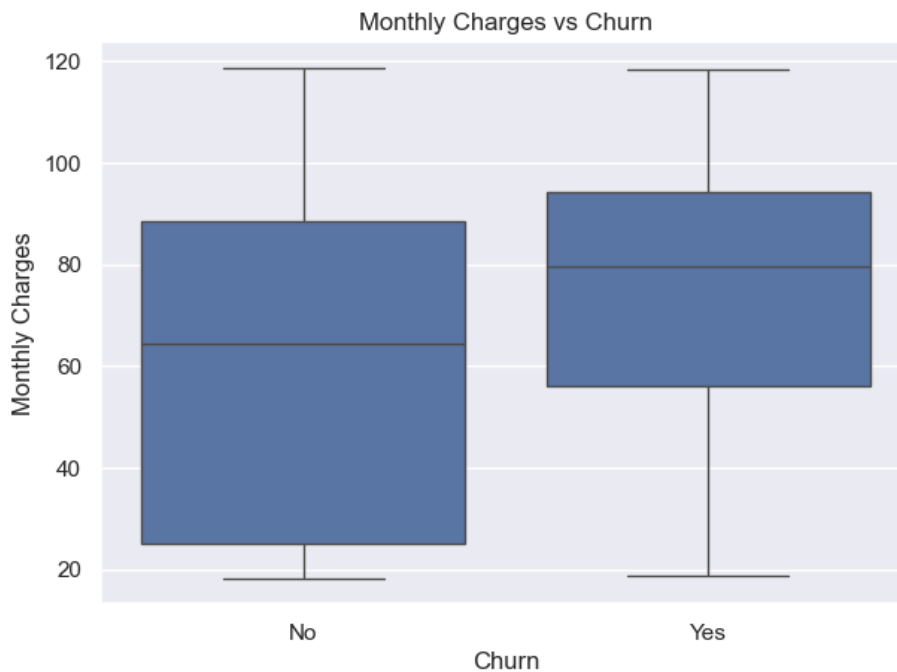
6. Visualizations



Observation: Most customers remained with the company, but approximately one out of every four customers churned, indicating a significant retention challenge



Observation: Customers with month-to-month contracts show the highest churn rate, while long-term contract customers are more loyal.



Observation: Customers with higher monthly charges are more likely to churn, suggesting pricing may influence customer attrition.

7. Conclusion

The customer churn analysis identified several important factors influencing customer churn in the telecom business. Customers with month-to-month contracts and higher monthly charges showed a higher tendency to churn. In contrast, customers with longer tenure were found to be more loyal and less likely to discontinue the service.

The analysis also highlighted the importance of customer retention strategies in reducing revenue loss and improving customer satisfaction. Additionally, the basic churn prediction model achieved an accuracy of 81.48%, demonstrating the potential of predictive analytics in identifying customers at risk of churn.

8. Recommendations

- Encourage customers to shift to long-term contracts.
- Provide loyalty rewards for long-tenure customers.
- Offer personalized discounts to high-risk customers.
- Improve customer support and service quality.
- Develop proactive customer retention campaigns.